CapExIQ — Complete Project Guide & Executive Q&A

Automated Micro-Fulfilment Centre Capital Budgeting Evaluation | NovaRetail GCC

Entity: NovaRetail GCC | Valuation: 6-Year DCF | WACC: 11.50% | Status: APPROVED

INITIAL OUTLAY	NET PRESENT VALUE	INTERNAL RETURN	PAYBACK PERIOD
AED 22.0M	AED 4.68M	17.65%	3.70 Yrs
Time Zero Outlay	Baseline NPV	Hurdle: 11.50%	Discounted: 4.68 Yrs

1. EXECUTIVE PROJECT SUMMARY

NovaRetail GCC is a major hypothetical omnichannel retail group operating across the UAE. Facing rapid e-commerce expansion in Dubai and Abu Dhabi, NovaRetail's manual warehouse fulfillment system has become an operational bottleneck, driving up order preparation costs (AED 14.50/order) and missing 2-hour delivery SLAs.

Management has evaluated an investment of **AED 22.0 Million** to construct an **Automated Micro-Fulfilment Centre (MFC)** equipped with Goods-to-Person (G2P) robotics, automated storage/retrieval systems (ASRS), and Warehouse Control System (WCS) integration in urban Dubai.

Executive Recommendation: APPROVE OPTION 1 (Full Investment)
The baseline model yields a Net Present Value of **AED 4,682,752**, an Internal Rate of Return of **17.65%** (exceeding the 11.50% WACC hurdle rate), a Profitability Index of **1.21x**, and an 88.4% probabilistic probability of success under 5,000 Monte Carlo simulation runs.

2. FINANCIAL RETURN & MODEL SCHEDULE

Metric Name	Calculated Value	Target Benchmark	Decision Status
Net Present Value (NPV)	**AED 4,682,752**	> AED 0	Passed (Approve)
Internal Rate of Return (IRR)	**17.65%**	> 11.50% WACC	Passed (+6.15% Spread)
Modified IRR (MIRR)	**14.28%**	> 11.50% Reinvestment	Passed (Economically Sound)
Profitability Index (PI)	**1.21x**	> 1.00x	Passed (1.21 Value Created per 1.00 Outlay)
Simple Payback Period	**3.70 Years**	< 4.0 Years	Passed (Recouped in Year 4)
Discounted Payback Period	**4.68 Years**	< 5.0 Years	Passed (Recouped in Year 5)

3. CAPITAL OUTLAY & CASH FLOW BREAKDOWN

- **Automation Equipment**: AED 18,000,000 (Goods-to-Person robotics, conveyor racks)
- **Installation & Systems Integration**: AED 2,500,000 (Mechanical, Electrical, WCS software connection)
- **Software & Cybersecurity**: AED 1,200,000 (WCS software licences and cybersecurity hardening)
- **Training & Launch Support**: AED 300,000 (Staff retraining)
- **Initial Working Capital**: AED 1,000,000 (Inventory buffer, 100% recovered at Year 6 end)
- **UAE Corporate Tax (9%)**: Deducted annually after straight-line equipment depreciation shields.

4. EXECUTIVE Q&A HANDBOOK (20 KEY QUESTIONS & ANSWERS)

Q1: Why is NovaRetail GCC investing in an automated micro-fulfilment centre?

NovaRetail GCC needs to serve fast-growing online grocery demand in Dubai and Abu Dhabi. Manual warehouse picking is too slow and expensive (costing AED 14.50 per order). Automation cuts order fulfillment cost to AED 4.20 and enables guaranteed 2-hour delivery.

Q2: How much money is needed up front, and how will it be spent?

The initial capital outlay is **AED 22.0 Million** in Year 0. This is spent on robotics hardware (AED 18M), integration (AED 2.5M), software (AED 1.2M), staff training (AED 300K), and working capital (AED 1.0M).

Q3: What is Net Present Value (NPV), and why is AED 4.68 Million good?

NPV measures how much net value the project adds to NovaRetail GCC today, after discounting all future cash flows back to present value using NovaRetail's 11.50% financing cost. An NPV of AED 4.68 Million means the project adds AED 4.68M in net shareholder wealth.

Q4: What is Internal Rate of Return (IRR) compared to the WACC hurdle rate?

IRR is the annualized return percentage earned by the investment. At **17.65%**, the project's return is 6.15 percentage points higher than NovaRetail's 11.50% cost of capital (WACC), proving it is highly profitable.

Q5: Why do we calculate Modified IRR (MIRR) in addition to standard IRR?

Standard IRR unrealistically assumes cash generated by the project is reinvested at 17.65%. MIRR conservatively assumes cash is reinvested at the company's realistic 11.50% hurdle rate. The project's MIRR of **14.28%** confirms it remains solid under conservative assumptions.

Q6: What is the Profitability Index (PI), and why is 1.21x important?

PI divides present value inflows by initial outlay. A PI of **1.21x** means that every 1.00 AED invested generates 1.21 AED in present value, proving efficient capital usage.

Q7: How long until NovaRetail GCC gets its money back?

The Simple Payback Period is **3.70 Years** (recouped during Year 4). The Discounted Payback Period (accounting for interest costs) is **4.68 Years** (recouped during Year 5).

Q8: What happens in a worst-case scenario (higher costs, lower demand)?

Under a severe stress test (+15% capex overrun, -25% benefit shortfall, 14.5% interest rate), NPV drops to negative AED -1.85M. CapExIQ automatically alerts management to switch to **Option 2 (Phased Pilot)** to cap initial risk at AED 12M.

Q9: How is UAE Corporate Tax (9%) included in the financial model?

Compliant with UAE Federal Tax law (effective June 2023), 9% tax is applied to operating profit (EBIT) after deducting annual equipment depreciation, reducing tax liability and giving accurate after-tax cash flows.

Q10: How are electricity costs calculated for operating the robots?

The model integrates real DEWA (Dubai Electricity & Water Authority) commercial tariffs (23-38 Fils/kWh plus 6.5 Fils fuel surcharge). Annual electricity consumption for 24/7 robot fleet operation is estimated at 650,000 kWh, costing ~AED 250,000 per year.

Q11: What is a Monte Carlo simulation, and what does it tell the Board?

Monte Carlo runs 5,000 randomized market scenarios altering costs, demand, labor savings, and interest rates. It proves the project has an **88.4% probability of achieving a positive NPV**, offering strong quantitative confidence.

Q12: What happens to working capital and equipment at Year 6 end?

The initial **AED 1.0M Working Capital** is 100% recovered in cash in Year 6. Equipment salvage value is conservatively estimated at **AED 1.5M**, giving a total Year 6 terminal cash inflow of AED 2.5M.

Q13: Which single variable has the biggest impact on success (Sensitivity Analysis)?

Tornado analysis shows **Annual Operating Cost Savings** is most sensitive. A 10% change in savings shifts project NPV by AED 3.8 Million, making operational execution the top priority.

Q14: How does the AI Assistant work, and can it make decisions automatically?

The AI Assistant acts strictly as an advisory financial officer, drafting explanations and answering questions. It cannot alter mathematical calculations or approve investments without human review.

Q15: Why is Full Investment recommended over Phased Investment?

Full Investment produces higher total NPV (AED 4.68M vs AED 2.10M) and unlocks maximum operational economies of scale immediately. Phased investment reduces initial outlay but adds AED 1.8M in re-mobilization costs.

Q16: How does the automated center improve daily retail operations?

It expands order throughput from 1,200 to **6,000 orders/day**, reduces order picking error rates from 2.4% to **0.2%**, and cuts order prep time from 45 minutes to **8 minutes**.

Q17: How was the 11.50% WACC hurdle rate calculated?

WACC combines 40% Debt (bank loan at 6.5% interest, post-tax 5.9%) and 60% Equity (required return of 14.5% based on UAE equity market risk premium). The weighted average is exactly 11.50%.

Q18: What are the main implementation risks during construction?

Key risks include software integration delays with legacy ERP, robotics shipping lead times, and staff retraining. These are mitigated through 6 stage-gate milestones and vendor penalty SLA clauses.

Q19: How does CapExIQ verify data accuracy and prevent errors?

CapExIQ features an automated CSV Data Quality Inspector that schema-validates all inputs for missing values, duplicates, and invalid numbers before passing data to the financial engine.

Q20: What is the final action for NovaRetail GCC's Board?

The Board should **APPROVE OPTION 1 (Full Investment of AED 22.0M)** with conditions: (1) Vendor SLA penalty clauses, (2) Strict AED 25.0M cost overrun ceiling, and (3) Quarterly stage-gate milestone audits.